

MULTI-STEP AI PROMPT CHAINING ASSIGNMENT

Scenario: Student Feedback to College Canteen Service Improvement Plan

Goal

The goal of this workflow is to analyze raw student feedback about a college canteen, identify the main service problems and student expectations, and transform those findings into a practical service improvement plan.

The workflow consists of three chained stages:

- Step 1 – Extraction and Planning: Analyze raw feedback and organize the major complaints, positive points, customer needs and service gaps.
- Step 2 – Generation and Drafting: Use the Step 1 analysis to prioritize the most important problem and create a targeted improvement plan.
- Step 3 – Evaluation and Refinement: Critically review the improvement plan against specific criteria and produce a polished final version.

Each step uses the output of the previous step, creating a continuous chained workflow.

STEP 1: EXTRACTION AND PLANNING

Objective

Analyze messy student feedback and identify recurring complaints, positive experiences, student expectations and possible service gaps. Organize the findings into a structured summary that can be used in the next step.

Prompt 1 – Feedback Extraction

Act as an expert student experience analyst.

I am going to provide you with messy and unorganized feedback about a college canteen called CampusBite Canteen.

Your task is to:

1. Identify the main student groups mentioned in the feedback.
2. Extract the major complaints and positive points.
3. Identify the main student needs and expectations.
4. Identify service strengths and weaknesses based only on the feedback.
5. Identify clear service gaps or unmet student needs.
6. Create a structured feedback summary that can be used to develop a service improvement plan in the next step.

Do not invent student experiences or canteen services that are not present in the feedback. Clearly separate observations from conclusions.

Raw Student Feedback:

- First-year students said the meals are affordable, but the queue becomes very long between classes.
- Final-year students liked the food variety, but sometimes popular items are unavailable during lunch.
- Some students appreciated the clean dining area, but said there are not enough tables during the busiest lunch

period.

- Students said they would like faster service during class breaks.
- Some students want clearer information when a popular food item is unavailable.
- Students generally like the reasonable prices, food variety and clean dining area.
- The canteen should improve busy-period service without increasing prices or reducing food quality.

AI RESPONSE – STEP 1

Structured Feedback Summary

1. Main Customer/User Groups

- First-year students looking for affordable meals and faster service between classes.
- Final-year students who value food variety and availability of popular items.
- Students who value clean dining areas and adequate seating.

2. Feedback Analysis

Area	Positive Point	Complaint / Limitation
Food	Affordable meals and good variety	Popular items can be unavailable during lunch
Service	Students appreciate the existing service	Queues become very long during class breaks
Seating	Clean dining area	Not enough tables during the busiest lunch period
Availability Communication	Students generally have a positive experience	Students want clearer information when items are unavailable

3. Main Needs and Expectations

- Faster service during class breaks and peak lunch periods.
- Clearer information when popular food items are unavailable.
- Better seating availability during the busiest lunch period.
- Maintaining reasonable prices and food variety.
- Maintaining a clean dining area and good food quality.

4. Strengths

- Reasonable prices.
- Food variety.
- Clean dining area.
- Students generally value the existing food quality.

5. Service Gaps

- Long queues during busy class-break periods.
- Popular food items may be unavailable during lunch.
- Limited table availability during peak lunch.
- Insufficient communication about unavailable items.

6. Initial Opportunity

The main opportunity is to improve peak-period service speed, food-availability communication and seating while preserving the canteen's existing strengths of reasonable prices, food variety and cleanliness.

STEP 2: GENERATION AND DRAFTING

Objective

Use the structured student feedback summary generated in Step 1 to identify the strongest service problem and create a targeted service improvement plan for CampusBite Canteen.

Prompt 2 – Improvement Plan Generation

Excellent. Now use only the structured student feedback summary you just created to develop a targeted service improvement plan for CampusBite Canteen.

Your task is to:

- 1. Select the most important unmet student need.
- 2. Identify the strongest service gap that CampusBite Canteen should address first.
- 3. Explain how the canteen can address that gap using only information supported by the feedback summary.
- 4. Create a clear three-part service improvement plan.
- 5. Create a short student-facing message describing the improvement.

Follow these constraints:

- Base the plan only on the student feedback summary.
- Do not claim new technologies, services or facilities that were not mentioned in the original feedback.
- Keep the recommendations practical and specific.
- Use simple and professional language.
- Focus on improving the student experience while preserving existing strengths.

AI RESPONSE – STEP 2

Targeted Gap

The strongest unmet need is faster and more predictable service during class breaks and peak lunch periods, especially for students with limited time between classes.

Strongest Problem

The main service gap is that queues become very long during busy periods, while popular food items may also be unavailable without clear information.

How the Organization Can Address the Gap

CampusBite Canteen can focus on faster peak-period service, clearer communication about food availability and better attention to table availability while maintaining reasonable prices, food variety and cleanliness.

Three-Part Improvement Plan

Priority Action	Expected Benefit
1. Peak-Period Queue Service — Focus staff attention on faster order handling during class breaks and busy	Reduced waiting time for students.

lunch periods.	
2. Food Availability Communication — Give students clearer information when popular items are unavailable.	Less uncertainty and frustration.
3. Peak Lunch Seating — Pay closer attention to table availability during the busiest lunch period.	Better seating experience for students.

Customer/User-Facing Message

CampusBite Canteen is improving busy-period service with faster queues, clearer food-availability information and better attention to seating—while keeping the reasonable prices, variety and clean environment students value.

STEP 3: CRITIQUE AND REFINEMENT

Objective

Evaluate the service improvement plan created in Step 2 against a specific checklist and refine it into a final version.

Prompt 3 – Critique and Refinement

Before we finalize the CampusBite Canteen service improvement plan, act as a strict senior student experience manager.

Review the service improvement plan and student-facing message you just created against this checklist:

1. Does the plan clearly address the main student groups identified in the feedback?
2. Does it directly address the strongest unmet need?
3. Does it focus on the strongest service gap rather than unrelated issues?
4. Does it use only information supported by the original feedback?
5. Are the recommendations practical and specific?
6. Does the plan preserve the canteen's existing strengths?
7. Is the student-facing message concise and clear?
8. Are there any unsupported claims or unnecessary statements?

First, provide a concise critique and identify any weaknesses.

Second, produce the final corrected service improvement plan and student-facing message.

Do not merely describe the corrections. Apply them to the final version.

AI RESPONSE – STEP 3

Senior Reviewer Critique

- The target student groups are reflected through the focus on first-year and final-year students and the broader student population.
- The plan addresses the main unmet need: faster service during peak periods.
- The plan also addresses food-availability communication and peak lunch seating.
- The recommendations remain within the information supported by the original feedback.
- The three priorities are practical and directly connected to the identified gaps.
- The plan preserves reasonable prices, food variety, food quality and cleanliness.
- The student-facing message is concise and can be made slightly more direct.

- The final version should clearly prioritize faster peak-period service, clearer food-availability communication and better attention to seating.

FINAL CORRECTED IMPROVEMENT PLAN

Priority 1: Improve Peak-Period Service

CampusBite Canteen should prioritize faster order handling during busy class-break and lunch periods to reduce student waiting time.

Priority 2: Improve Food Availability Communication

When a popular food item is unavailable, students should receive clearer information so they know what choices are available.

Priority 3: Improve Peak Lunch Seating Management

The canteen should pay closer attention to table availability during the busiest lunch period to improve the seating experience.

Final Customer/User-Facing Message

CampusBite Canteen is improving peak-period service with faster order handling, clearer food-availability information and better attention to seating—while keeping the reasonable prices, food variety and clean environment students value.

WORKFLOW SUMMARY

Step	Input	Output
Step 1 – Extraction/Planning	Raw student feedback	Structured student needs, strengths, weaknesses and service-gap analysis
Step 2 – Generation/Drafting	Step 1 feedback summary	Targeted service gap, improvement plan and student-facing message
Step 3 – Evaluation/Refinement	Step 2 improvement plan	Critique + final corrected improvement plan

Chaining Dependency

Step 2 explicitly uses the structured student feedback summary generated in Step 1 to identify the strongest service gap and create a targeted improvement plan. Step 3 explicitly reviews the improvement plan generated in Step 2 against a defined checklist and refines it. Therefore, the workflow demonstrates a genuine multi-step chained prompting process rather than three independent prompts.

CONCLUSION

This workflow demonstrates how chained AI prompts can transform raw student feedback into a practical college canteen service improvement plan. The first prompt extracts student groups, complaints, strengths and needs, the second prompt transforms those findings into a targeted improvement plan, and the third prompt evaluates the result against defined criteria and produces a refined version. The workflow demonstrates the required progression: Extraction/Planning → Generation/Drafting → Evaluation/Refinement.

SUBMISSION EVIDENCE TO ADD

Before submitting, run the three prompts in the assigned AI interface in the same chat session and add screenshots or the direct chat link showing the live conversation. The screenshots should clearly show Prompt 1 and its response, Prompt 2 and its response, and Prompt 3 and its response.

[Add live chat screenshots or direct chat link here.]